

C.1. Devrenin DC analizi yapılırsa,

$$V_{CC} = R_B \cdot I_B + V_{BE} + (R_{E1} + R_{E2}) \cdot I_E$$

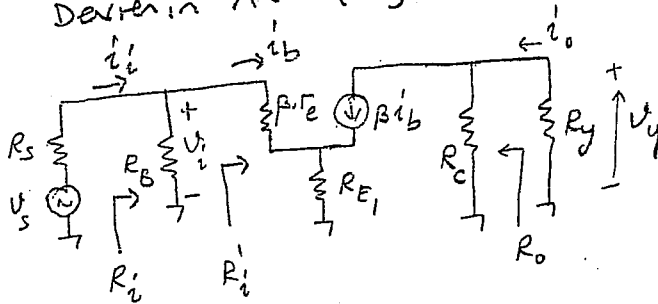
$$V_{CC} - V_{BE} = I_B (R_B + R_{E1} \cdot (1 + \beta))$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + R_{E1} \cdot (1 + \beta)} = \frac{22 - 0,7}{330k + 1,67k \cdot 81} = \frac{21,3}{330k + 1,67k \cdot 81}$$

$$I_B = \frac{21,3}{465,27} = 0,046 \text{ mA} \quad I_E = I_B (1 + \beta) = 0,046 \times 81 = 3,73 \text{ mA}$$

$$r_e = \frac{V_T}{I_E} = \frac{26 \text{ mV}}{3,73 \text{ mA}} = 6,97 \Omega$$

Devrenin AC eşdeğer modeli çizilecek olursa.



$$a) \quad v_y = + \beta i_b \cdot R_C // R_L$$

$$v_i = \beta r_e \cdot i_b + (1 + \beta) i_b \cdot R_{E1}$$

$$v_i = i_b (\beta r_e + (1 + \beta) R_{E1})$$

$$A_v = \frac{v_y}{v_i} = - \frac{\beta \cdot i_b \cdot R_C // R_L}{i_b \cdot (\beta r_e + \beta R_{E1})}$$

$$A_{v_s} = \frac{v_y}{v_s} = \frac{v_y}{v_i} \cdot \frac{v_i}{v_s} = A_v \cdot \frac{R_i}{R_s + R_i}$$

$$A_v = - \frac{R_C // R_L}{r_e + R_{E1}}$$

$$R_i' = \frac{v_i}{i_b} = \beta r_e + (1 + \beta) R_{E1} = \frac{80 \cdot 6,97 \Omega}{0,56k} + \frac{81 \cdot 0,47k}{38,07k} = 38,63k$$

$$A_v = - \frac{R_C // R_L}{r_e + R_{E1}} = \frac{0,85k}{6,97 \Omega + 0,47k} = \frac{0,85k}{0,477k} = 1,78$$

$$R_i = R_B // R_i' = 330k // 38,63 = 34,58k$$

$$A_{v_s} = 1,78 \cdot \frac{34,58k}{1k + 34,58k} = 1,73$$

C.d. b. $i_o = \beta i_b + \frac{v_y}{R_c} = \beta i_b + \frac{-\beta i_b \cdot R_c // R_y}{R_c}$

$$i_o = \beta i_b \left[1 - \frac{R_c // R_y}{R_c} \right] = \beta i_b \left[1 - \frac{R_c \cdot R_y}{R_c (R_c + R_y)} \right]$$

$$i_o = \beta i_b \left[1 - \frac{R_y}{R_c + R_y} \right] = \beta i_b \left[\frac{R_c + R_y - R_y}{R_c + R_y} \right]$$

$$i_o = \beta i_b \cdot \frac{R_c}{R_c + R_y} \quad \text{--- (1)}$$

$$i_i \cdot R_i = i_b \cdot \beta r_e + (1 + \beta) i_b \cdot R_{E1}$$

$$i_i \cdot \frac{R_B \cdot R_i'}{R_B + R_i'} = i_b [\beta r_e + (1 + \beta) R_{E1}]$$

$$i_i \cdot \frac{R_B \cdot [\beta r_e + (1 + \beta) R_{E1}]}{R_B + [\beta r_e + (1 + \beta) R_{E1}]} = i_b \cdot [\beta r_e + (1 + \beta) R_{E1}]$$

$$i_b = \frac{R_B}{R_B + [\beta r_e + (1 + \beta) R_{E1}]} \cdot i_i$$

②. → ① de yazılırsa,

$$A_i = \frac{i_o}{i_i} = \frac{R_c}{R_c + R_y} \cdot \beta \cdot \frac{R_B}{R_B + \beta r_e + (1 + \beta) R_{E1}} = \frac{5,6k}{5,6k + 1k} \cdot 80 \cdot \frac{330k}{330k + 0,56k + 38,07k}$$

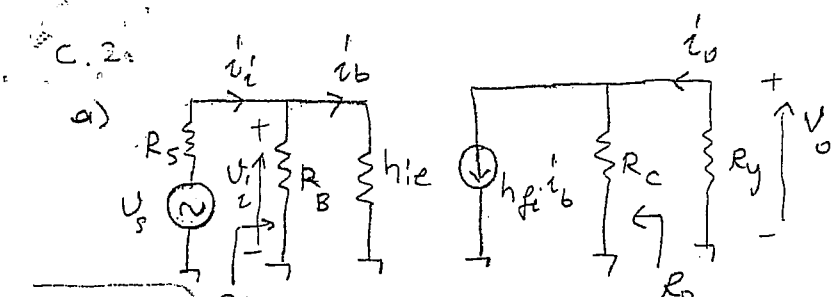
$$A_i = 60,87$$

$$A_i = \frac{R_i}{R_y} |A_v| = \frac{34,58k}{1k} \cdot 1,78 = 61,55$$

c) $R_i = 34,58k$

d) $R_o = 5,6k$

$$v_i = \frac{v_b + i_b R_B}{R_B}$$



$$V_o = -h_{fe} \cdot i_b \cdot R_C \parallel R_L$$

$$V_i = h_{ie} \cdot i_b$$

$$A_V = \frac{V_o}{V_i} = -\frac{h_{fe} \cdot R_C \parallel R_L}{h_{ie}}$$

$$A_V = -\frac{110}{0,83k} \cdot \frac{3k \cdot 4,7k}{3k + 4,7k} = -242,7$$

$$R_i = R_B \parallel h_{ie}$$

$$R_i = 470k \parallel 0,83k$$

$$R_i \approx 0,83k$$

$$A_{VS} = A_V \cdot \frac{R_i}{R_s + R_i} = 242,7 \cdot \frac{0,83k}{0,6k + 0,83k}$$

$$A_{VS} \approx 140,9$$

b)

$$i_i = \frac{V_i}{R_B} + i_b, \quad V_i = h_{fe} \cdot i_b, \quad \Rightarrow i_i = \frac{h_{fe} \cdot i_b}{R_B} + i_b$$

$$i_i = i_b \left(\frac{h_{fe}}{R_B} + 1 \right) = i_b \left(\frac{h_{fe} + R_B}{R_B} \right) \Rightarrow i_b = \frac{R_B}{h_{fe} + R_B} \cdot i_i \quad \text{--- (1)}$$

$$i_o = h_{fe} \cdot i_b + \frac{V_o}{R_C} = h_{fe} \cdot i_b + \frac{-h_{fe} \cdot i_b \cdot R_C \parallel R_L}{R_C}$$

$$i_o = i_b \cdot h_{fe} \left(1 - \frac{R_C \cdot R_L}{R_C(R_C + R_L)} \right) = i_b \cdot h_{fe} \cdot \left(1 - \frac{R_L}{R_C + R_L} \right)$$

$$i_o = i_b \cdot h_{fe} \left(\frac{R_C + R_L - R_L}{R_C + R_L} \right) = i_b \cdot h_{fe} \left(\frac{R_C}{R_C + R_L} \right) \quad \text{--- (2)}$$

① → ② de gauru sa,

$$A_i = \frac{i_o}{i_i} = \frac{R_B}{h_{fe} + R_B} \cdot h_{fe} \cdot \frac{R_C}{R_C + R_L} = \frac{470k}{0,83k + 470k} \cdot 110 \cdot \frac{3k}{3k + 4,7k}$$

$$A_i = 42,78$$

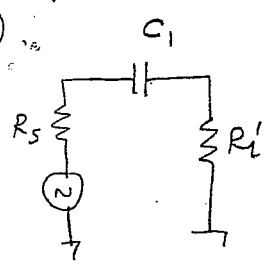
$$\frac{2,427}{A_i} = \frac{R_i}{R_L} \cdot |A_V| = \frac{0,83k}{4,7k} \cdot 242,7 = 42,86$$

c)

$$R_i = R_B \parallel h_{ie} = 470k \parallel 0,83k \approx 0,83k$$

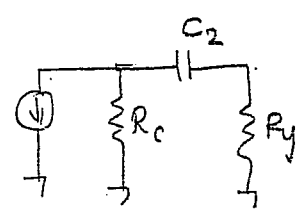
$$R_o = R_C = 3k$$

d)



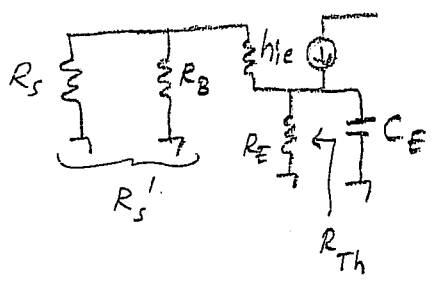
$$f_{11} = \frac{1}{2\pi C_1 (R_S + R'_i)} = \frac{1}{2\pi \cdot 1 \cdot 10^{-6} \cdot (0,6k + 0,83k)}$$

$$f_{11} = 111,3 \text{ Hz}$$



$$f_{12} = \frac{1}{2\pi C_2 \cdot (R_c + R_y)} = \frac{1}{2\pi \cdot 1 \cdot 10^{-6} \cdot (3k + 4,7k)}$$

$$f_{12} = 20,7 \text{ Hz}$$



$$R_e = R_{Th} = \frac{R'_S + h_{ie}}{h_{fe}} \parallel R_E$$

$$R_e = \frac{0,6k \parallel 470k + 0,83k}{110} \parallel 0,9k$$

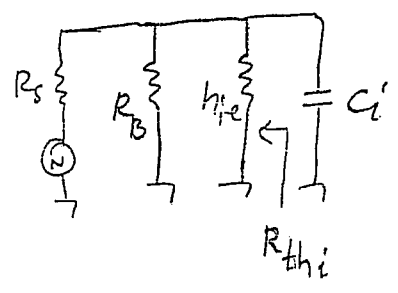
$$R_e = \frac{0,6k + 0,83k}{110} \parallel 0,9k$$

$$R_e = 13,7 \Omega \parallel 900 \Omega \approx 13 \Omega$$

$$f_{13} = \frac{1}{2\pi R_e \cdot C_E} = \frac{1}{2\pi \cdot 13 \cdot 10 \cdot 10^{-6}} = 1224 \text{ Hz}$$

At kesim frekansı, $f_1 = f_{13} = 1224 \text{ Hz}$ olur.

e)



$$R_{Thi} = R_S \parallel R_B \parallel h_{ie} = 0,6k \parallel 470k \parallel 0,83k = 0,35k$$

$$C_i = C_{w_i} + C_{be} + C_{m_i}$$

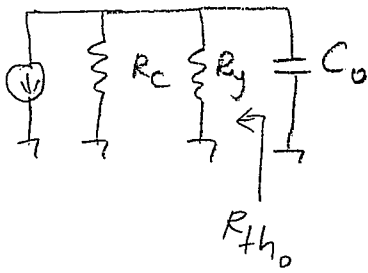
$$C_{m_i} = C_{bc} (1 - A_v) = 6pF (1 - (-242,7)) = 1462,2pF$$

$$C_i = 7pF + 20pF + 1462,2pF = 1489,2pF$$

$$f_{21} = \frac{1}{2\pi R_{Thi} \cdot C_i} = \frac{1}{2\pi \cdot 0,35 \cdot 10^3 \cdot 1489,2 \cdot 10^{-12}} = 305,35 \text{ kHz}$$

C.2. e. devamı.

sy.5.



$$R_{Th_0} = R_C // R_Y = 3k // 4,7k = 1,83k$$

$$C_0 = C_{w_0} + C_{ce} + C_{m_0} = 11pF + 10pF + 6,03pF = 27,03pF$$

$$C_{m_0} = C_{bc} \left(1 - \frac{1}{A_V}\right) = 6pF \left(1 - \frac{1}{-242,7}\right)$$

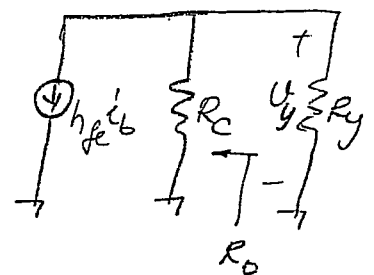
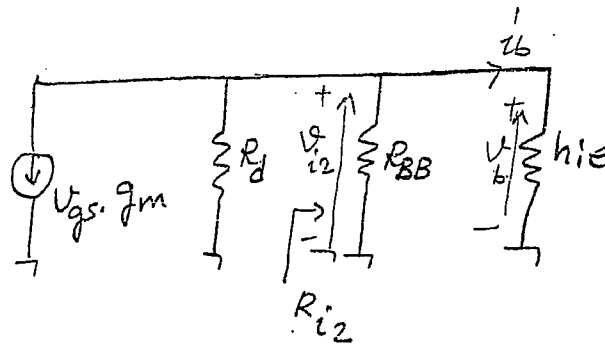
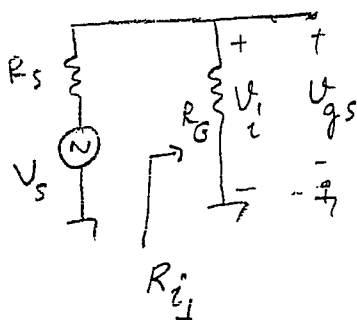
$$C_{m_0} = 6,025pF$$

$$f_{22} = \frac{1}{2\pi R_{Th_0} C_0} = \frac{1}{2\pi \cdot 1,83 \cdot 10^3 \cdot 27,03 \cdot 10^{-12}} = 3217,5kHz$$

üst kesim frekansı olarak,

$$f_2 = f_u = 305,35kHz \text{ alınır}$$

C.3.



a) $R_{i1} = R_G = 1M$, $R_{BB} = 24k // 8,2k = 6,11k$,

$$R_{i2} = R_{BB} // h_{ie} = 6,11k // 1,2k = 1k$$

$$R_0 = R_C = 2,7k$$

b) $A_{V1} = \frac{V_{i2}}{V_i} = - \frac{V_{g_s} \cdot g_m \cdot R_d // R_{i2}}{V_{g_s}} = - g_m \cdot R_d // R_{i2} = -2,5 \cdot 1,8k // 1k$

$$A_{V1} = -1,61$$

$$A_{V2} = \frac{V_y}{V_{i2}} = - \frac{h_{fe} \cdot i_b \cdot R_C // R_Y}{h_{ie} \cdot i_b} = - \frac{h_{fe}}{h_{ie}} \cdot R_C // R_Y = - \frac{150}{1,2k} \cdot 2,7k // 1k$$

$$A_{V2} = -91,2$$

$$A_V = \frac{V_y}{V_i} = \frac{V_g}{V_{i2}} \cdot \frac{V_{i2}}{V_i} = A_2 \cdot A_1 = (-1,61)(-91,2) = 146,8$$

C.8. b. devamı.

Sy.6.

$$A_{vs} = \frac{v_y}{v_s} = \frac{v_y}{v_i'} \cdot \frac{v_i'}{v_s} = A_v \cdot \frac{R_{i1}}{R_s + R_{i1}} = 146,8 \cdot \frac{1M}{18k + 1M} = 144,2$$

$$C.) v_y = A_{vs} \cdot v_s = 144,2 \times 1mV = 144,2 mV$$